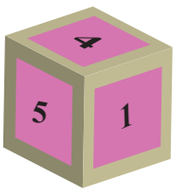
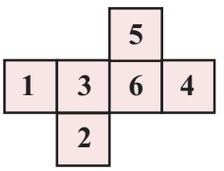
Example for equally likely events: Suppose a fair die is rolled.

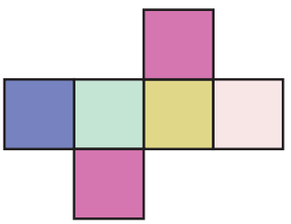
Number on the face	1	2	3	4	5	6
Chance of occurrence	1	1	1	1	1	1





Example for not equally likely events: A colour die is shown in figure is rolled.

Colour on the face					
Chance of occurrence	1	1	1	2	1





Similarly, suppose if we toss a coin, the events of getting a head or a tail are equally likely.

Methods to find sample space

Illustration 12.5

Two coins are tossed, the sample space is

(i) $S = \{H, T\} \times \{H, T\} = \{(H, H), (H, T), (T, H), (T, T)\}$ or $\{HH, HT, TH, TT\}$

(ii) If a coin is tossed and a die is rolled simultaneously, then the sample space is

$S = \{H, T\} \times \{1, 2, 3, 4, 5, 6\} = \{H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6\}$ or

$S = \{(H, 1), (H, 2), (H, 3), (H, 4), (H, 5), (H, 6), (T, 1), (T, 2), (T, 3), (T, 4), (T, 5), (T, 6)\}$.

Also one can interchange the order of outcomes of coin and die. The following table gives the sample spaces for some random experiments.

Random Experiment	Total Number of Outcomes	Sample space
Tossing a fair coin	$2^1 = 2$	$\{H,\ T\}$
Tossing two coins	$2^2 = 4$	$\{HH,\ HT,\ TH,\ TT\}$
Tossing three coins	$2^3 = 8$	$\{HHH,\ HHT,\ HTH,\ THH,\ HTT,\ THT,\ TTH,\ TTT\}$
Rolling fair die	$6^1 = 6$	{1, 2,3, 4, 5,6}
Rolling Two dice or single die two times.	$6^2 = 36$	{(1,1) ,(1,2), (1,3), (1,4), (1,5), (1,6), (2,1), (2,2), (2,3), (2,4), (2,5), (2,6), (3,1), (3,2), (3,3), (3,4), (3,5), (3,6), (4,1), (4,2), (4,3), (4,4), (4,5), (4,6), (5,1), (5,2), (5,3), (5,4), (5,5), (5,6), (6,1), (6,2),(6,3), (6,4), (6,5), (6,6)}
Drawing a card from a pack of 52 playing cards	$52^1 = 52$	Heart ♥ A 2 3 4 5 6 7 8 9 10 J Q K Red in colour Diamond ♦ A 2 3 4 5 6 7 8 9 10 J Q K Red in colour Spade ♠ A 2 3 4 5 6 7 8 9 10 J Q K Black in colour Club ♣ A 2 3 4 5 6 7 8 9 10 J Q K Black in colour

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